MSG No. 129.420 P_c4/nmm [Type IV, tetragonal]

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

No.	tag	matrix (polar)	det	TR
1	$\{1\|0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
2	$\{4^+_{001}\|\frac{1}{2}00\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{2} \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
3	$\{4^-_{001}\|0\frac{1}{2}0\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
4	$\{2_{100}\|\frac{1}{2}00\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	1	1
5	$\{2_{010}\|0\frac{1}{2}0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & 0 \end{bmatrix}$	1	1
6	$\{2_{001}\|\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
7	$\{2_{110}\|\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{2} \\ 1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & 0 \end{bmatrix}$	1	1
8	$\{2_{1-10}\|0\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	1	1
9	$\{-1\|0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	−1	1
10	$\{-4^+_{001}\|\frac{1}{2}00\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{2} \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	−1	1
11	$\{-4^-_{001}\|0\frac{1}{2}0\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & 0 \end{bmatrix}$	−1	1
12	$\{\mathrm{m}_{100}\|\frac{1}{2}00\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	−1	1
13	$\{\mathrm{m}_{010}\|0\frac{1}{2}0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	−1	1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
14	$\{\mathrm{m}_{001}\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}1&0&0&\frac{1}{2}\\0&1&0&\frac{1}{2}\\0&0&-1&0\end{bmatrix}$	-1	1
15	$\{\mathrm{m}_{110}\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}0&-1&0&\frac{1}{2}\\-1&0&0&\frac{1}{2}\\0&0&1&0\end{bmatrix}$	-1	1
16	$\{\mathrm{m}_{1-10}\vert 0\}$	$\begin{bmatrix}0&1&0&0\\1&0&0&0\\0&0&1&0\end{bmatrix}$	-1	1
17	$\{1'\vert 00\frac{1}{2}\}$	$\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&\frac{1}{2}\end{bmatrix}$	1	-1
18	$\{4^+_{001}{}'\vert\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix}0&-1&0&\frac{1}{2}\\1&0&0&0\\0&0&1&\frac{1}{2}\end{bmatrix}$	1	-1
19	$\{4^-_{001}{}'\vert 0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}0&1&0&0\\-1&0&0&\frac{1}{2}\\0&0&1&\frac{1}{2}\end{bmatrix}$	1	-1
20	$\{2_{100}{}'\vert\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix}1&0&0&\frac{1}{2}\\0&-1&0&0\\0&0&-1&\frac{1}{2}\end{bmatrix}$	1	-1
21	$\{2_{010}{}'\vert 0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}-1&0&0&0\\0&1&0&\frac{1}{2}\\0&0&-1&\frac{1}{2}\end{bmatrix}$	1	-1
22	$\{2_{001}{}'\vert\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&-1&0&\frac{1}{2}\\0&0&1&\frac{1}{2}\end{bmatrix}$	1	-1
23	$\{2_{110}{}'\vert\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}0&1&0&\frac{1}{2}\\1&0&0&\frac{1}{2}\\0&0&-1&\frac{1}{2}\end{bmatrix}$	1	-1
24	$\{2_{1-10}{}'\vert 00\frac{1}{2}\}$	$\begin{bmatrix}0&-1&0&0\\-1&0&0&0\\0&0&-1&\frac{1}{2}\end{bmatrix}$	1	-1
25	$\{-1'\vert 00\frac{1}{2}\}$	$\begin{bmatrix}-1&0&0&0\\0&-1&0&0\\0&0&-1&\frac{1}{2}\end{bmatrix}$	-1	-1
26	$\{-4^+_{001}{}'\vert\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix}0&1&0&\frac{1}{2}\\-1&0&0&0\\0&0&-1&\frac{1}{2}\end{bmatrix}$	-1	-1
27	$\{-4^-_{001}{}'\vert 0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}0&-1&0&0\\1&0&0&\frac{1}{2}\\0&0&-1&\frac{1}{2}\end{bmatrix}$	-1	-1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
28	$\{\mathrm{m}_{100}{}'\vert\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	-1	-1
29	$\{\mathrm{m}_{010}{}'\vert0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	-1	-1
30	$\{\mathrm{m}_{001}{}'\vert\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	-1	-1
31	$\{\mathrm{m}_{110}{}'\vert\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{2} \\ -1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	-1	-1
32	$\{\mathrm{m}_{1-10}{}'\vert00\frac{1}{2}\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	-1	-1